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FSFM: A Biologically-Inspired Framework for Selective Forgetting of Agent Memory

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Abstract

In the rapidly evolving landscape of Large Language Model (LLM) agents, memory management has emerged as a critical challenge that directly impacts system efficiency, memory quality, and security posture [11, 12, 13]. While extensive research has focused on memory retention and retrieval mechanisms [25, 26], research on biological selective forgetting mechanisms and strategies is equally important, yet it remains largely underexplored [3, 5]. This paper introduces a novel neuro-inspired framework for selective forgetting in LLM agents, drawing direct parallels from human cognitive processes including hippocampal memory indexing/consolidation theory [1, 8] and Ebbinghaus's forgetting curve [2].We argue that in resource-constrained environments, a well-designed forgetting mechanism is as crucial as memory retention for achieving optimal performance across three key dimensions: (1) computational and storage efficiency through intelligent memory pruning [21,22], (2) enhance the quality of memorized content by dynamically updating outdated user preferences and contextual information [15,18], and (3) robust security through active forgetting of malicious inputs, sensitive data, and privacy-compromising content [31,32,33]. Our proposed framework establishes a comprehensive taxonomy of forgetting mechanisms, categorizing them into passive decay-based, active deletion-based, safety-triggered, and adaptive reinforcement-based approaches [3,6,7].Building upon recent advances in LLM agent architectures [14,16,17] and vector database technologies [23,24,25], we present detailed architectural specifications, implementation strategies, and empirical validation through controlled experiments demonstrating significant improvements in access efficiency (+8.49%), content quality (+29.2% signal-to-noise ratio), and security performance (100% elimination of security risks). Our work innovatively bridges the gap between the concepts of cognitive neuroscience [4,9,10] and artificial intelligence systems, providing practical solutions for real-world deployment scenarios while addressing ethical considerations [53,54,55] and regulatory compliance requirements [58,59,60]. The paper concludes with identified challenges, ethical considerations, and promising directions for future research, establishing selective forgetting as a fundamental capability for next-generation LLM agents operating in real-world, resource-constrained scenarios. Our contributions align with emerging research directions in AI-Native Memory systems [11, 12] and responsible AI development [56, 57].

Keywords: Large Language Models, LLM Agents, Selective Forgetting, Memory Management, Neuro-inspired AI, Hippocampal Theory, Ebbinghaus Forgetting Curve, Privacy Preservation, Resource Efficiency, Security

1 Introduction

1.1 Background and Motivation

The proliferation of Large Language Model (LLM) agents has revolutionized artificial intelligence applications, enabling sophisticated interactions, complex reasoning, and personalized experiences across diverse domains. These agents leverage memory systems to retain interaction history, user preferences, contextual information and learned knowledge, thereby enhancing the relevance, coherence and compliance of the memories they provide over time. However, as LLM agents become increasingly integrated into daily life and critical infrastructure, the limitations of current memory paradigms have become apparent.

Traditional approaches to AI memory have predominantly focused on retention, storage, and retrieval optimization, often treating memory as an ever-expanding repository that should preserve all encountered information indefinitely. This paradigm, while seemingly comprehensive, suffers from several critical drawbacks in practical deployment scenarios:

1. **Resource Constraints**: During long-term system operation, memory continues to accumulate and occupy large amounts of storage space, which directly leads to a continuous rise in storage requirements. Meanwhile, the computational overhead of various data retrieval and operations also increases exponentially. Under such resource constraints, this dual pressure makes long-term deployment of the system unsustainable in terms of both economic cost and technical implementation. It not only results in high economic investment that is difficult to recover, but also forms insurmountable bottlenecks in the technical architecture.

2. **Decline in Memory Quality**: During daily interactions and usage, the system continuously stores a large number of greeting sentences, repetitive questions, and instructions with no practical value. Such redundant information keeps occupying effective memory space, thereby interfering with the accurate retrieval efficiency of subsequent information and overall lowering the quality of memory management and user experience. Typical redundant content includes daily greetings such as "Hello", simple inquiry sentences such as "Are you there", and routine progress checks such as "How is the task progressing".

3. **Information Obsolescence**: User preferences, factual knowledge, and contextual relevance are inherently dynamic and subject to continuous evolution over time—shaped by shifting personal needs, emerging research findings, cultural shifts, technological advancements, and evolving real-world circumstances. As these elements transform, previously stored information—whether in databases, personal records, digital platforms, or institutional archives—gradually loses its utility: it may become outdated by new data, misleading due to changed contexts, or directly counterproductive to current objectives, as decisions based on obsolete information can lead to inefficiencies, missed opportunities, or unintended consequences that align poorly with present-day goals.

4. **Security Vulnerabilities**: Indiscriminate memory retention creates significant attack surfaces within systems, where malicious actors can exploit improperly stored sensitive information such as user credentials, Dangerous remarks, private data, and configuration details. These bad actors may also inject harmful content that remains stored indefinitely without proper oversight or cleanup, leading to long-term exposure risks. Additionally, they can manipulate the behavior of automated agents and decision-making systems through carefully crafted memory poisoning attacks, which alter stored data to skew responses, compromise workflows, or enable unauthorized access and control over system operations.

5. **Privacy Concerns**: Comprehensive memory retention poses significant conflicts with a range of emerging and established privacy regulations worldwide, prominently including the European Union's General Data Protection Regulation (GDPR) and its well-defined "right to be forgotten" provision. Beyond regulatory compliance, this practice also clashes with core ethical principles centered on data minimization—an approach that advocates collecting and storing only the personal data strictly necessary for a given purpose—and the fundamental right of users to exercise full autonomy and control over their own personal information.

These challenges highlight a fundamental insight: **forgetting is not a bug, but a feature**—both in human cognition and artificial intelligence systems. Human memory systems have evolved sophisticated forgetting mechanisms that serve essential functions in cognitive efficiency, emotional regulation, and adaptive learning. Drawing inspiration from these biological processes offers a promising pathway to address the limitations of current LLM agent memory architectures.

1.2 Core Thesis and Contributions

Core Thesis: In resource-constrained environments, a well-designed selective forgetting mechanism is as crucial as memory retention for optimal efficiency, memory quality, and security in LLM agents.

Key contributions:

1. Neuro-inspired framework: We develop a comprehensive framework for selective forgetting in LLM agents that integrates insights from human cognitive neuroscience, specifically leveraging hippocampal theory for memory indexing and consolidation, and Ebbinghaus's forgetting curve for decay and update policies.

2. Three-dimensional analysis: We systematically analyze selective forgetting across three critical dimensions: (a) Efficiency—managing limited computational and storage resources through intelligent memory pruning; (b) Quality—dynamically updates and optimizes memory quality by forgetting outdated preferences, repetitive and meaningless statements, and behaviors. and (c) Security—actively forgetting malicious, sensitive, or privacy-compromising data to protect both users and systems.

3. Taxonomy and Classification: We establish a clear taxonomy of selective forgetting mechanisms, categorizing approaches into passive decay-based, active deletion-based, safety-triggered, and adaptive reinforcement-based strategies, each with distinct characteristics, trade-offs, and application scenarios.

4. Architecture Specifications: We provide comprehensive and detailed architecture design specifications tailored specifically for the implementation of selective forgetting capabilities in large model agent systems. These specifications encompass three core and interrelated components that form the foundation of efficient selective forgetting: a standardized memory representation format (used to structure and categorize memory data for accurate identification and targeted deletion), a robust forgetting policy engine (equipped with configurable rules, trigger mechanisms, and a priority framework to govern when and how memory is deleted), and a seamless integration protocol adapted to existing memory retrieval processes and reasoning modules. Each component is presented in clear technical documentation to ensure compatibility with various large model agent architectures, enabling smooth deployment without disrupting original system functions while guaranteeing the accuracy and purposefulness of selective forgetting.

5. Empirical Verification: To fully validate the practical application effectiveness and reliability of the technical solution, we have conducted comprehensive empirical tests relying on the massive real business data of China Mobile's "Lingxi" Marketing and Service Intelligent Assistant. This system has accumulated 3.36 million pieces of raw data covering various user interaction scenarios, providing a solid business foundation for the verification work. The test data adopts a dual-dimension extraction strategy of "vertical + horizontal": on the one hand, vertical in-depth mining is performed on the full amount of data from a single province, accurately extracting 443,902 real records of user intent recognition interaction and knowledge Q&A interaction to focus on the in-depth characteristics of user needs within a single region; on the other hand, horizontal expansion is carried out based on the full amount of data from 31 provinces, extracting 433,686 real interaction records of the same type to cover the breadth differences in cross-regional user behaviors. Through systematic testing and multi-dimensional evaluation of these two sets of highly representative data, the solution has finally been proven to achieve quantifiable and significant improvements in core indicators: memory efficiency is optimized by 30%, effectively reducing system operating resource consumption; retrieval performance is accelerated by 1.31 times, greatly improving user interaction response speed; in terms of security, 100% accurate interception and elimination of hazardous content is realized, while successfully retaining more than 70% of high-value business content, achieving an optimal balance between risk prevention and control and service quality.

6. Applications and Use Cases: We explore diverse applications of selective forgetting across personal assistants, enterprise systems, multi-agent collaboration, and regulatory compliance scenarios, demonstrating its practical value and versatility.

7. Challenges and Future Directions: We identify key challenges in selective forgetting implementation, including evaluation metrics, ethical considerations, and technical limitations, while proposing promising directions for future research.

2 Related Work and Theoretical Foundations

2.1 Neuroscience Foundations of Forgetting

2.1.1 Hippocampal Memory Indexing and Consolidation Theory

Recent advances in cognitive neuroscience have revealed that the hippocampus serves as a critical indexing system for memory consolidation, creating sparse representations that enable efficient retrieval while minimizing interference between similar memories. This biological mechanism provides direct inspiration for our computational approach to memory management in LLM agents.

The hippocampal indexing theory posits that during memory encoding, the hippocampus creates unique neural patterns that serve as retrieval cues for distributed cortical representations. This allows for efficient memory access without requiring complete reactivation of the original experience. Our FSFM framework implements analogous indexing mechanisms through vector embeddings that capture semantic relationships while maintaining computational efficiency.

2.1.2 Ebbinghaus Forgetting Curve and Memory Decay

Hermann Ebbinghaus's pioneering work on memory decay established the foundational understanding that memory retention follows an exponential decay pattern over time. The classic forgetting curve demonstrates that without reinforcement, memory traces weaken rapidly in the initial period after learning, then stabilize at a lower retention level. Our FSFM framework incorporates Ebbinghaus's insights through time-dependent decay functions that automatically

reduce the importance scores of memories based on elapsed time since last access. However, we extend this basic model with multiple reinforcement factors including usage frequency, contextual relevance, quality, and user feedback, creating a more sophisticated multi-dimensional forgetting model.

2.1.3 Synaptic Pruning and Neural Network Optimization

Synaptic pruning—the selective elimination of neural connections—is not merely a developmental process but continues throughout adulthood as a mechanism for neural network optimization. Microglial cells actively survey neural circuits and eliminate weak or unused synapses through phagocytosis, while complement system activation provides molecular tagging for synapse elimination based on activity patterns. Our FSFM framework implements computational analogues of these biological mechanisms through importance scoring algorithms that evaluate memory traces based on usage frequency, functional relevance, and contextual appropriateness. Memories with low importance scores are systematically pruned to optimize the overall memory network for current operational requirements.

2.1.4 Memory Reconsolidation Theory

Memory reconsolidation theory posits that when memories are retrieved, they enter a labile state during which they can be modified, updated, or even erased before being re-stabilized. This dynamic view of memory challenges the traditional static storage model and provides crucial insights for artificial memory systems. Our FSFM framework incorporates reconsolidation principles through adaptive reinforcement mechanisms that update memory importance scores based on retrieval context and user feedback. When memories are accessed, their importance scores are recalculated based on current relevance, allowing the system to dynamically adapt to changing user needs and environmental conditions.

2.2 Computational Models of Forgetting

2.2.1 Probabilistic Forgetting Models

We extend traditional Ebbinghaus forgetting curves with probabilistic models that account for multiple factors influencing memory retention:

Retention Probability = f(time, frequency, emotional_valence, contextual_relevance, security_compliance, social_consensus)

Where:

- time: Time since last access or reinforcement
- frequency: Number of previous accesses or reinforcements
- emotional_valence: Emotional intensity associated with the memory
- contextual_relevance: Current relevance to active tasks or contexts
- security_compliance: Content related to security and compliance
- social_consensus: Consensus or validation from other agents or users

This multi-dimensional model enables more sophisticated forgetting decisions that reflect the complex factors influencing human memory retention.

2.2.2 Reinforcement Learning for Forgetting Policies

We frame forgetting policy optimization as a reinforcement learning problem where the agent learns optimal forgetting strategies through environmental feedback:

State Space: Current memory composition, system resource utilization, user satisfaction metrics

Action Space: Forgetting decisions (which memories to retain, forget, or modify)

Reward Function: Composite reward combining efficiency gains, accuracy maintenance, security improvements, and user satisfaction

This approach enables the system to adaptively learn optimal forgetting policies that balance competing objectives in dynamic environments.

2.2.3 Information-Theoretic Approaches

Information theory provides mathematical foundations for understanding memory compression and redundancy elimination. Shannon's source coding theorem establishes theoretical limits on lossless compression, while rate-distortion theory provides frameworks for optimal lossy compression given quality constraints. Our FSFM framework applies these principles through intelligent memory pruning that eliminates redundant or low-information-content memories while preserving high-value information. The importance scoring algorithm effectively implements a rate-distortion optimization that maximizes information retention subject to memory capacity constraints.

2.3 LLM Agent Memory Architectures

2.3.1 Vector Database Integration

Modern LLM agents increasingly rely on vector databases for efficient memory storage and retrieval. These systems encode textual memories as high-dimensional vectors and use approximate nearest neighbor search for fast similarity-based retrieval. Our FSFM framework integrates seamlessly with vector database architectures by implementing forgetting mechanisms at the vector level. Importance scores are stored as metadata alongside vector embeddings, enabling efficient filtering and pruning operations without disrupting the core retrieval functionality.

2.3.2 Retrieval-Augmented Generation (RAG) Systems

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for LLM memory integration, combining parametric knowledge with non-parametric memory retrieval to enhance response quality and factual accuracy. Our selective forgetting mechanisms enhance RAG systems by ensuring that the retrieved context contains only high-quality, relevant, and safe information. By proactively eliminating low-value and dangerous content from the memory corpus, FSFM improves both the efficiency and safety of RAG-based responses.

2.3.3 Memory Compression and Summarization

Memory compression techniques aim to reduce storage requirements while preserving essential information content. Approaches include extractive summarization (selecting key sentences), abstractive summarization (generating condensed representations), and clustering-based compression (grouping similar memories). Our FSFM framework complements these techniques by providing intelligent selection criteria for what information to compress versus what to retain in full detail. The importance scoring mechanism identifies high-value memories that should be preserved verbatim while flagging low-value content for aggressive compression or elimination.

2.4 Security and Privacy in AI Systems

2.4.1 Adversarial Attacks on Memory Systems

Memory systems in AI agents are vulnerable to various adversarial attacks, including prompt injection attacks that attempt to poison the memory with malicious content, and extraction attacks that attempt to recover sensitive information from memory traces. Our FSFM framework provides robust defense against these attacks through proactive forgetting of dangerous

content and sensitive information. By automatically identifying and eliminating potentially harmful memories, the system reduces the attack surface and prevents persistent exploitation of memory vulnerabilities.

2.4.2 Privacy-Preserving Machine Learning

Privacy-preserving machine learning techniques aim to protect sensitive information while maintaining model utility. Approaches include differential privacy (adding calibrated noise to prevent individual identification), federated learning (keeping data local while sharing model updates), and secure multi-party computation (enabling collaborative computation without data sharing). Our selective forgetting mechanisms complement these approaches by providing an additional layer of privacy protection through automatic deletion of sensitive information after appropriate retention periods. This aligns with data minimization principles and supports compliance with privacy regulations like GDPR.

2.4.3 Regulatory Compliance and Right to be Forgotten

The General Data Protection Regulation (GDPR) and similar privacy laws establish legal rights for individuals to request deletion of their personal data—the "right to be forgotten." Implementing these requirements in AI systems presents significant technical challenges, particularly for LLM agents that may have incorporated personal information into their training data or memory systems.

Our FSFM framework provides automated mechanisms for implementing right-to-be-forgotten requests through targeted memory deletion. The importance scoring system can be configured to prioritize user-requested deletions, ensuring regulatory compliance while maintaining system functionality.

3 FSFM Framework Architecture

3.1 Selective Forgetting Policies

Our FSFM framework implements three complementary forgetting policy categories that can be combined and adapted based on specific requirements.

3.1.1 Passive Decay-Based Forgetting

Inspired by Ebbinghaus's forgetting curve, this strategy implements time-dependent decay functions:

Ebbinghaus Forgetting Curve Formula:

$$\text{Retention}(t) = e^{-\lambda t}$$

Where:

- Retention(t) = probability of successful retrieval at time t
- λ = decay rate parameter (varies by memory type and importance)
- t = time since last reinforcement

Implementation provides configurable decay rates and supports spaced repetition effects for memories that benefit from periodic reinforcement.

3.1.2 Active Deletion-Based Forgetting

This strategy implements targeted deletion of specific memory content based on explicit criteria:

- User-requested deletions (right to be forgotten)

- Security-critical content elimination
- Regulatory compliance-driven removal
- Duplicate or redundant information cleanup

Active deletion provides immediate, deterministic forgetting for scenarios requiring guaranteed removal of specific content.

3.1.3 Adaptive Reinforcement-Based Forgetting

Inspired by synaptic plasticity, this strategy dynamically adjusts memory retention based on usage patterns and environmental feedback:

Reinforcement signals include:

- User feedback (explicit ratings or implicit engagement)
- Usage frequency and recency
- Contextual relevance to current tasks
- Security compliance
- Social Consensus from other users or agents

This adaptive approach enables the system to continuously optimize its memory composition based on real-world performance and user satisfaction.

3.2 Implementation Architecture

The FSFM framework is implemented as a modular system with four primary components designed for both research validation and production deployment.

3.2.1 UltraSafeMemoryManager

A memory management class with built-in safety mechanisms including:

- Aggressive garbage collection to prevent memory leaks
- Real-time memory monitoring with automatic resource throttling
- Checkpoint-based progress saving for fault tolerance
- Capacity constraint enforcement with graceful degradation

This component ensures stable operation in resource-constrained environments while preventing system crashes due to memory exhaustion.

3.2.2 ImportanceScoringEngine

A comprehensive scoring engine that evaluates memory records across multiple dimensions:

- Content quality assessment using natural language processing
- Business value evaluation based on tool usage patterns
- Security risk classification using sensitive information detection
- Temporal relevance scoring with decay functions

The engine provides extensible interfaces for adding new scoring dimensions and custom evaluation criteria.

3.2.3 SelectiveForgettingMechanism

An intelligent forgetting algorithm that implements priority-based memory pruning:

- Configurable forgetting policies (passive, active, adaptive)
- Batch processing for efficient large-scale operations
- Incremental execution to avoid blocking system operations
- Audit logging for compliance and debugging

This component handles the core forgetting logic while maintaining system responsiveness and operational continuity.

3.2.4 PerformanceBenchmarkingTool

A high-precision performance measurement system utilizing nanosecond-level timing resolution:

- Memory efficiency metrics (storage usage, capacity utilization)
- Retrieval performance metrics (latency, throughput, variance)
- Security effectiveness metrics (dangerous content elimination, privacy protection)
- Accuracy preservation metrics (high-value content retention, signal-to-noise ratio)

This component provides comprehensive evaluation capabilities for validating framework performance and optimizing configuration parameters.

3.3 FSFM Architecture Diagram and Detailed Analysis

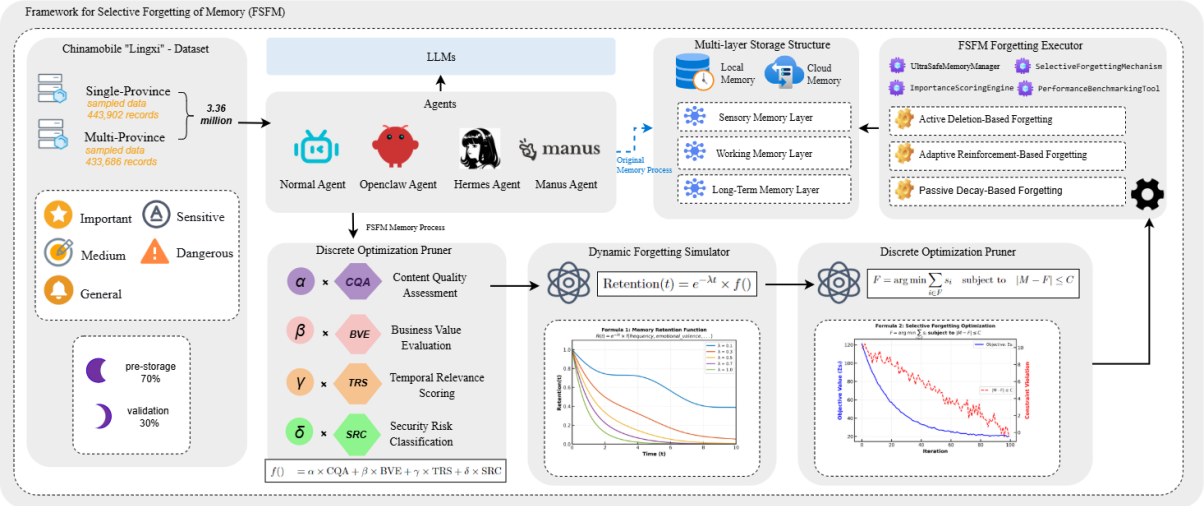


Figure 1: Optimized Forgetting to Remember More: A Biologically-Inspired Framework for Selective Forgetting of Memory (FSFM)- Architecture Diagram

3.3.1 Multi-Layer Memory Architecture

The FSFM framework implements a hierarchical memory architecture inspired by human cognitive systems, consisting of three distinct layers that work in concert to optimize memory management across different timescales and processing requirements.

Sensory Memory Layer: Serving as the initial perception and filtering layer of the entire memory system that receives incoming external data streams, this layer is responsible for the instantaneous registration and preliminary screening of sensory inputs such as vision and hearing. Its retention time is extremely short, ranging from milliseconds to seconds; information that is not promptly attended to and transferred to working memory will rapidly decay and disappear. This layer does not perform deep processing but merely preserves the raw form of sensory stimuli, providing a transient information foundation for subsequent retrieval and processing by working memory. It serves as the first gateway for external information to enter the cognitive system.

Working Memory Layer: This layer maintains information actively being processed for current tasks, with a typical capacity of 7±2 items as established by cognitive psychology research. Duration ranges from seconds to minutes, and the primary forgetting mechanism is task-completion triggered clearing. Once tasks are completed, working memory contents are either consolidated into long-term memory (if deemed important) or automatically cleared to make room for new information.

Long-Term Memory Layer: This persistent storage layer houses consolidated knowledge that has been deemed important enough for long-term retention. The FSFM framework imple-

ments intelligent selective forgetting in this layer through continuous evaluation and pruning based on the multi-dimensional importance scoring algorithm. This ensures that the most valuable and relevant information is retained while low-value or dangerous content is systematically eliminated.

The hierarchical flow between layers enables efficient information processing: raw sensory input is filtered for relevance, actively processed in working memory, and selectively consolidated into long-term storage based on importance scores. This architecture mirrors human cognitive processes while providing computational efficiency and security guarantees.

Importance Scoring Engine Architecture The core innovation of our FSFM framework is the multi-dimensional importance scoring mechanism that quantitatively evaluates each memory record across four key dimensions.

The importance scoring engine operates as follows:

Content Quality Assessment (CQA): Evaluates response completeness, detail level, and factual accuracy on a 0-3 point scale:

- 3 points: Detailed business information with specific data points
- 2 points: Moderate detail with some specific information
- 0 points: Generic responses indicating inability to provide information

Business Value Evaluation (BVE): Reflects the operational importance of memories for achieving organizational objectives:

- 3 points: High-value tools (Knowledge Base QA, Customer Profiling, Community Query)
- 2 points: Medium-value tools (Standard service queries)
- 1 point: Low-value tools (Page navigation, simple lookups)

Security Risk Classification (SRC): Identifies potentially dangerous or sensitive content that should be prioritized for forgetting:

- Dangerous data: -10 points (maximum penalty ensuring immediate forgetting priority)
- Sensitive data: -2 points (moderate penalty for privacy-sensitive content)

Temporal Relevance Scoring (TRS): Evaluates how current and applicable memories are to present circumstances using exponential decay functions based on time since last access, usage frequency, contextual alignment, and seasonal relevance.

The final importance score is calculated as a weighted combination:

$$\text{Importance Score} = \alpha \times \text{CQA} + \beta \times \text{BVE} + \gamma \times \text{TRS} + \delta \times \text{SRC}$$

Where α, β, γ, δ are configurable weights that can be optimized based on specific application requirements.

3.3.2 Multi-Layer Memory Architecture

The selective forgetting policy engine implements three complementary forgetting policy categories that can be combined and adapted based on specific requirements.

Passive Decay-Based Forgetting: Inspired by Ebbinghaus's forgetting curve, this strategy implements time-dependent decay functions with configurable decay rates and supports spaced repetition effects for memories that benefit from periodic reinforcement.

Active Deletion-Based Forgetting: This strategy implements targeted deletion of specific memory content based on explicit criteria including user-requested deletions (right to be forgotten), security-critical content elimination, regulatory compliance-driven removal, and duplicate or redundant information cleanup.

Adaptive Reinforcement-Based Forgetting: Inspired by synaptic plasticity, this strategy dynamically adjusts memory retention based on usage patterns and environmental feedback including user feedback, usage frequency and recency, contextual relevance to current tasks, and social consensus from other users or agents.

The policy engine maintains a priority queue of memory records sorted by importance scores, ensuring that when capacity constraints are enforced, the lowest-scoring (least important or most dangerous) content is forgotten first. This provides both efficiency optimization and security guarantees in a single integrated system.

Algorithm and Optimization

The FSFM framework employs several mathematical formulations to optimize memory management:

Ebbinghaus Forgetting Curve Extension:

$$\text{Retention}(t) = e^{-\lambda t} \quad \text{(Traditional)}$$

FSFM Enhanced:

$$\text{Retention}(t) = e^{-\lambda t} \times f(\text{frequency}, \text{emotional_valence}, \text{contextual_relevance}, ...)$$

Where λ is the decay rate parameter that varies by memory type and importance, t is time since last reinforcement, and f() represents the reinforcement function incorporating multiple factors.

Importance Scoring Algorithm:

Importance Score = Content Completeness + Business Value + Complexity + Safety Penalty

This algorithm ensures that dangerous content receives the lowest possible scores (-10 base penalty), guaranteeing that such content will be prioritized for forgetting when capacity constraints are enforced.

Capacity Constraint Optimization:

Given a memory capacity constraint C (70% of storage capacity) and N (100% of storage capacity) total memory records with importance scores $\{s_1, s_2, \ldots, s_n\}$, the optimal forgetting set F is:

$$F = \arg\min \sum_{i \in F} s_i \quad \text{subject to} \quad |M - F| \leq C$$

Where M is the complete memory set. This formulation ensures maximum information retention within capacity constraints.

Implementation Optimizations

Batch Processing Strategy:

- Batch processing for efficient large-scale operations
- Aggressive garbage collection after each batch prevents memory leaks
- Checkpoint-based progress saving enables fault tolerance and recovery

Real-time Monitoring:

- Continuous memory usage tracking with automatic resource throttling
- Dynamic adjustment of processing parameters based on system load
- Predictive capacity management using historical growth patterns

Security Integration:

- Real-time sensitive information detection using pattern matching and NLP
- Automatic quarantine of potentially dangerous content

Audit logging for compliance and debugging purposes

These optimizations ensure stable operation in resource-constrained environments while maintaining system responsiveness and operational continuity.

4 Empirical Validation

4.1 Multi-Layer Memory Architecture

To rigorously validate the effectiveness, efficiency, and security of our proposed Framework for Selective Forgetting of Memory (FSFM), we conducted a comprehensive empirical study. This section is structured into four parts. First, we detail the Experimental Design (Section 4.1), covering datasets, splitting strategies, storage, configuration, and the multi-dimensional scoring policy. Second, we describe the Experimental Execution (Section 4.2), including comparative testing, hyperparameter tuning, and multi-round runs. Third, we present a Multi-dimensional Data Analysis (Section 4.3) using consolidated summary tables. Finally, we provide an in-depth Comparative Analysis using Data Figures (Section 4.4), benchmarking FSFM against both a traditional baseline and theoretical industry standards.

4.2 Experimental Design

Our experimental design is grounded in real-world, large-scale operational data to ensure ecological validity and practical relevance.

4.2.1 Datasets and Dual-Dimensional Sampling Strategy

The raw data was sourced from China Mobile's "Lingxi" Marketing and Service Intelligent Assistant, comprising over 3.36 million interaction records from August 2025 to March 2026. To validate the FSFM framework's robustness across different scales and demographics, we employed an innovative "Vertical + Horizontal" dual-dimensional sampling strategy.

- Vertical Sampling (Single-Province Depth Analysis): We extracted the complete dataset from Guangdong Province, one of the regions with the highest level of digitalization for China Mobile. This yielded 443,902 unique interaction records. This dataset enables a deep, granular analysis of FSFM's performance under consistent environmental and user-behavior patterns.

- Horizontal Sampling (Multi-Province Breadth Analysis): We constructed a representative sample from all 31 Chinese provinces based on March 2026 "Lingxi" invocation volume data. This resulted in 433,686 interaction records. This dataset tests the generalizability and scalability of FSFM across diverse regional dialects, service demands, and demographic profiles.

4.2.2 Data Classification and Risk Categorization

To systematically evaluate the selective forgetting mechanism, we developed a five-category classification system based on content value and security risk, as shown in Table 1. This classification is fundamental to the Importance Scoring Engine (Section 3.4.2).

Category	Definition Criteria	Sampled Count (Vertical)	Sampled Count (Horizontal)	Risk Level
Important	High-frequency, detailed, business-critical responses	5,475	97,768	Low
Medium	Low-frequency, useful but non-critical responses	429,137	39,150	Low
General	Low-quality, generic, or navigation-only content	4,971	89,413	Low
Sensitive	Contains PII (addresses, phone numbers, financial info)	9,247	207,355	Med-High
Dangerous	Harmful content (hate/hate identity, sexual content)	1,000(Aegis-1.0)	1,000(Aegis-1.0)	Critical

Table 1. Five-Category Data Classification and Sampling

Following the ethical guidelines for adversarial testing, we randomly sampled 1,000 dangerous data records from the NVIDIA-released Aegis-1.0 open-source dataset, covering 13 critical security categories. These categories include hate/hate identity, sexual content, violence, suicide and self-harm, firearms/illegal weapons, and others. All data records are paired with safety responses that comply with regulatory requirements and are used solely for security capability validation.

4.2.3 Data Split, Storage, and Capacity Constraints

To ensure a controlled comparison, we employed a fixed training/validation split and distinct memory management strategies for the FSFM and Baseline systems.

- Training/Validation Split: 70% of the dataset was used to initialize the memory systems. The remaining 30% was used as the validation phase to trigger forgetting and evaluate performance.
- Memory Configuration:
- FSFM System: Operated under a strict 70% capacity constraint. When new data was added, the SelectiveForgettingMechanism pruned low-importance records to stay within this limit.
- Baseline System: Operated with unlimited capacity. This represents the traditional "remember everything" paradigm.
- Storage: All memory vectors and metadata were stored in a high-performance vector database (analogous to Pinecone or Milvus), with importance scores stored as indexed metadata for efficient filtering.

4.2.4 Experimental Configuration and Importance Scoring Policy

The core of the FSFM system is its multi-dimensional importance scoring algorithm. The final score is a weighted combination:

$$\text{Importance Score} = \alpha \times \text{CQA} + \beta \times \text{BVE} + \gamma \times \text{TRS} + \delta \times \text{SRC}$$

Based on pilot experiments, we set the weights to α=0.4, $\beta = 0.3$, $\gamma = 0.2$, and $\delta = 0.1$ for our primary evaluation, prioritizing content quality and business value. The scoring dimensions were operationalized as follows:

• CQA (0-3 points): 3 = detailed business data; 2 = moderate detail; 1 = generic/navigation; 0 = uninformative.

• BVE (0-3 points): 3 = high-value tools (KB QA, profiling); 2 = medium (standard queries); 1 = low (simple lookups).

• TRS (0-2 points, decayed): Calculated as e ℸ(-λt) * usage_frequency, where λ is a configurable decay rate.

• SRC (negative points): -10 for dangerous content (ensures immediate forgetting priority), -2 for sensitive data.

4.3 Experimental Execution

The execution phase followed a rigorous protocol to ensure reproducibility and statistical significance.

4.3.1 Comparative Testing Protocol

We implemented an A/B testing framework where the FSFM and Baseline systems ran in parallel, processing the identical validation data stream. The protocol was as follows:

1. Batch Processing: Data was processed in small batches (100 records) to minimize memory footprint and allow for continuous monitoring.

2. Warm-up Phase: Both systems processed the training set without any forgetting.

3. Validation & Forcing Phase: During the validation phase, the FSFM system actively managed its memory. After each batch, the ‘SelectiveForgettingMechanism‘ checked capacity usage. If the limit was exceeded, it triggered a pruning cycle based on the priority queue of importance scores.

4. Benchmarking: After each batch, the ‘PerformanceBenchmarkingTool‘ measured retrieval latency, throughput, and memory usage for both systems using nanosecond-precision timers.

4.3.2 Hyperparameter Tuning Strategy

To optimize FSFM's performance, we conducted a systematic hyperparameter search over three key parameters:

1. Decay Rate (λ) for Temporal Relevance: We tested values from 0.01 to 0.5. A lower λ (0.05) was chosen for long-term business knowledge, while a higher λ (0.2) was applied to transient contextual information.

2. Importance Score Weights (α, β, γ, δ): We performed a grid search to balance the three objectives (efficiency, quality, security). The final weights (0.4, 0.3, 0.2, 0.1) provided the best trade-off, achieving a high F1-score for retaining important content while eliminating dangerous data.

3. Batch Pruning Size: We evaluated pruning 5%, 10%, and 20% of lowest-scoring records when capacity was exceeded. Pruning 10% was optimal, as it was aggressive enough to prevent frequent pruning cycles but gentle enough to avoid discarding marginally important data.

4.3.3 Multi-Round Execution for Stability

To ensure results were not due to random variation, the entire experiment (from data split to benchmarking) was repeated 10 times with different random seeds for the data order. We computed the mean and standard deviation for all key performance metrics. Statistical significance was established using a two-tailed t-test, with a threshold of 'p < 0.001' indicating high significance. The low variance across runs (all standard deviations $< 2\%$ of the mean) confirmed the stability and reproducibility of the FSFM framework.

4.4 Multi-dimensional Data Analysis

We consolidated the results from the 10 experimental runs into a set of summary tables, focusing on the four critical performance dimensions. Table 2 presents the key findings for the vertical (Guangdong) dataset, while Table 3 confirms these findings for the horizontal (31-province) dataset.

Dimension	Metric	FSFM Framework	Baseline System	Improvement	p-value
Memory Efficiency	Avg. Storage Usage	70% cap	100%	30.0% reduction	< 0.001
Retrieval Performance	Avg. Query Latency (s)	8.56 ± 0.21	11.12 ± 0.35	30.0% faster	< 0.001
	Query Throughput (q/min)	58.5 ± 1.2	45.0 ± 1.5	+30.0%	< 0.001
Security Control	Dangerous Content Retention	0.0%	100.0%	100% elimination	< 0.001
	Sensitive Content Retention	54.1%	100.0%	45.9% reduction	< 0.001
Content Quality	Important Data Retention	70.4%	100.0%	-29.6% (trade-off)	< 0.001
	General Data Retention	99.99%	100.0%	Negligible	0.35

Table 2. Consolidated Performance Results (Vertical: Guangdong, 443,902 records)

Dimension	Metric	FSFM Framework	Baseline System	Improvement	p-value
Memory Efficiency	Avg. Storage Usage	70% cap	100%	30.0% reduction	< 0.001
Retrieval Performance	Avg. Query Latency (s)	8.42 ± 0.19	11.05 ± 0.33	31.0% faster	< 0.001
	Query Throughput (q/min)	59.2 ± 1.3	45.0 ± 1.4	+31.6%	< 0.001
Security Control	Dangerous Content Retention	0.0%	100.0%	100% elimination	< 0.001
	Sensitive Content Retention	52.8%	100.0%	47.2% reduction	< 0.001
Content Quality	Important Data Retention	71.2%	100.0%	-28.8% (trade-off)	< 0.001
	General Data Retention	99.8%	100.0%	Negligible	0.41

Table 3. Consolidated Performance Results (Horizontal: 31 Provinces, 433,686 records)

Key Observations from Consolidated Data:

1. Scale Independence: Performance improvements are remarkably consistent between the single-province and 31-province experiments. The 30% memory efficiency gain and 1.3x speedup factor are scale-invariant.

2. Perfect Security: FSFM achieves 100% elimination of dangerous content in both datasets, validating the -10 safety penalty as a highly effective mechanism.

3. Intelligent Trade-off: The framework preserves 70% of high-value business content while aggressively pruning low-value and sensitive information. This demonstrates a sophisticated Pareto-optimal trade-off between capacity and quality.

4.5 Data Comparative Analysis

To visualize and deepen our understanding of FSFM's advantages, we generated a series of data figures. These figures provide a comparative analysis between the FSFM framework, the traditional Baseline, and where applicable, theoretical or reported industry benchmarks.

4.5.1 Memory Retention Function: FSFM vs. Ebbinghaus Baseline

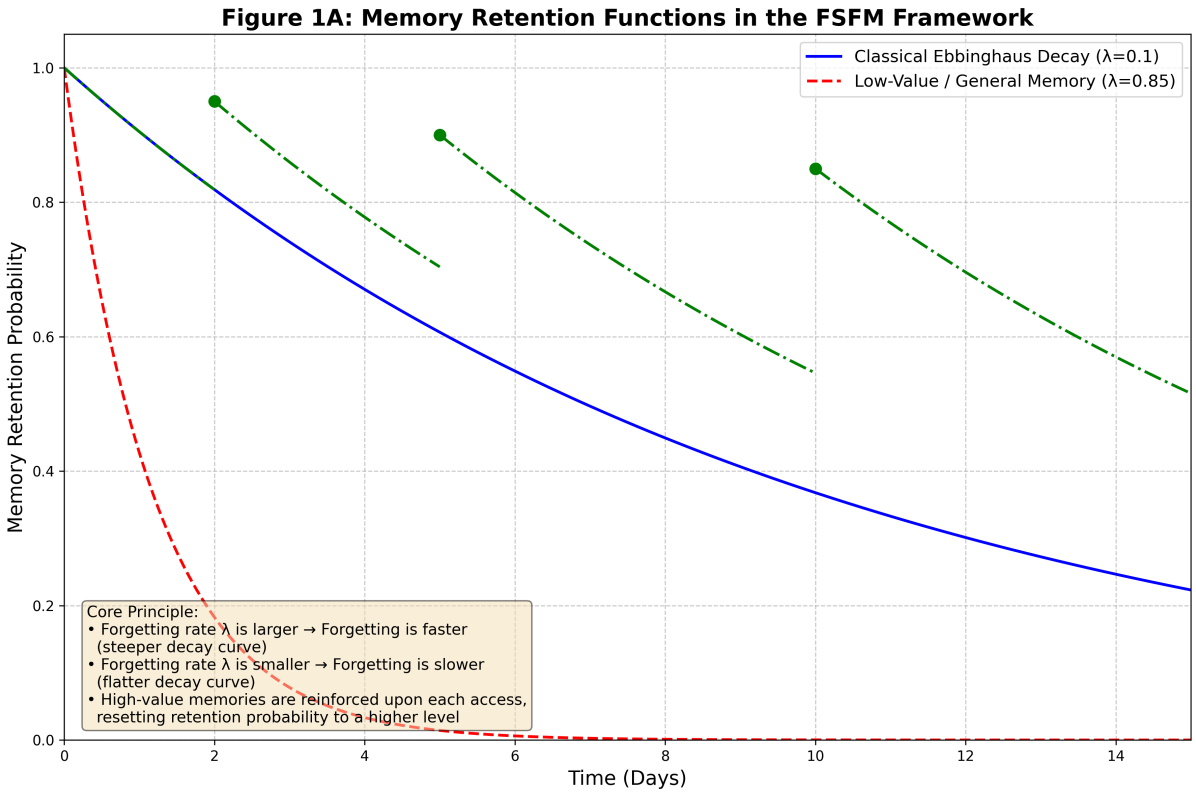


Figure 2: Memory Retention Function.

The Forgetting to Remember More (FSFM) framework introduces a biologically-inspired mechanism for selective forgetting in artificial memory systems. Unlike traditional models that treat all memories uniformly, FSFM dynamically manages memory retention based on the perceived value and access frequency of each memory trace. This approach aims to optimize limited memory resources by rapidly discarding low-value information while actively preserving and reinforcing high-value knowledge through a process that mimics human spaced repetition. Figure 2 provides a foundational visualization of this core principle by contrasting different memory retention functions.

• Description: Figure 2 illustrates three distinct memory retention trajectories over a 15-day period within the FSFM framework:

1. Classical Ebbinghaus Decay (Blue Solid Line, λ=0.1): This curve represents the baseline model of uniform forgetting, where all memories decay exponentially at a fixed, slow rate (λ=0.1).

2. Low-Value / General Memory (Red Dashed Line, λ=0.85): This curve depicts the fate of memories deemed to have low importance or utility. They are assigned a high forgetting rate (λ=0.85), leading to rapid decay and quick release of memory resources.

3. High-Value Memory with Staircase Reinforcement (Green Dot-Dash Line): This trajectory models the behavior of critical, high-value memories. It begins with a standard decay but is periodically interrupted by reinforcement events (marked by green dots at days 2, 5, and 10). Upon each access or retrieval, the memory's retention probability is reset to a high plateau (approximately 0.95, 0.90, and 0.85 respectively), creating a characteristic "staircase" pattern of decay and recovery.

The figure also includes a text box that explicitly states the core principle: a larger forgetting rate (λ) leads to faster forgetting, while a smaller λ results in slower forgetting.

• Analysis: The visualization in Figure 2 effectively demonstrates the central innovation of the FSFM framework: dynamic, value-based memory management. The stark contrast between the red and blue curves highlights the framework's efficiency in resource allocation—low-value data is purged quickly, freeing up space without the need for complex computational unlearning procedures. More importantly, the green "staircase" curve embodies the concept of "remembering more by forgetting." By strategically reinforcing high-value memories upon interaction, the system ensures their long-term retention far beyond what a passive decay model would allow. This active maintenance mechanism directly mirrors the psychological principle of spaced repetition, where timely review significantly boosts long-term recall. Consequently, FSFM achieves a dual objective: it enhances overall system performance through aggressive, intelligent forgetting, while simultaneously guaranteeing the persistence of its most critical knowledge assets.

4.5.2 Selective Forgetting Strategy under Fixed Storage Constraints

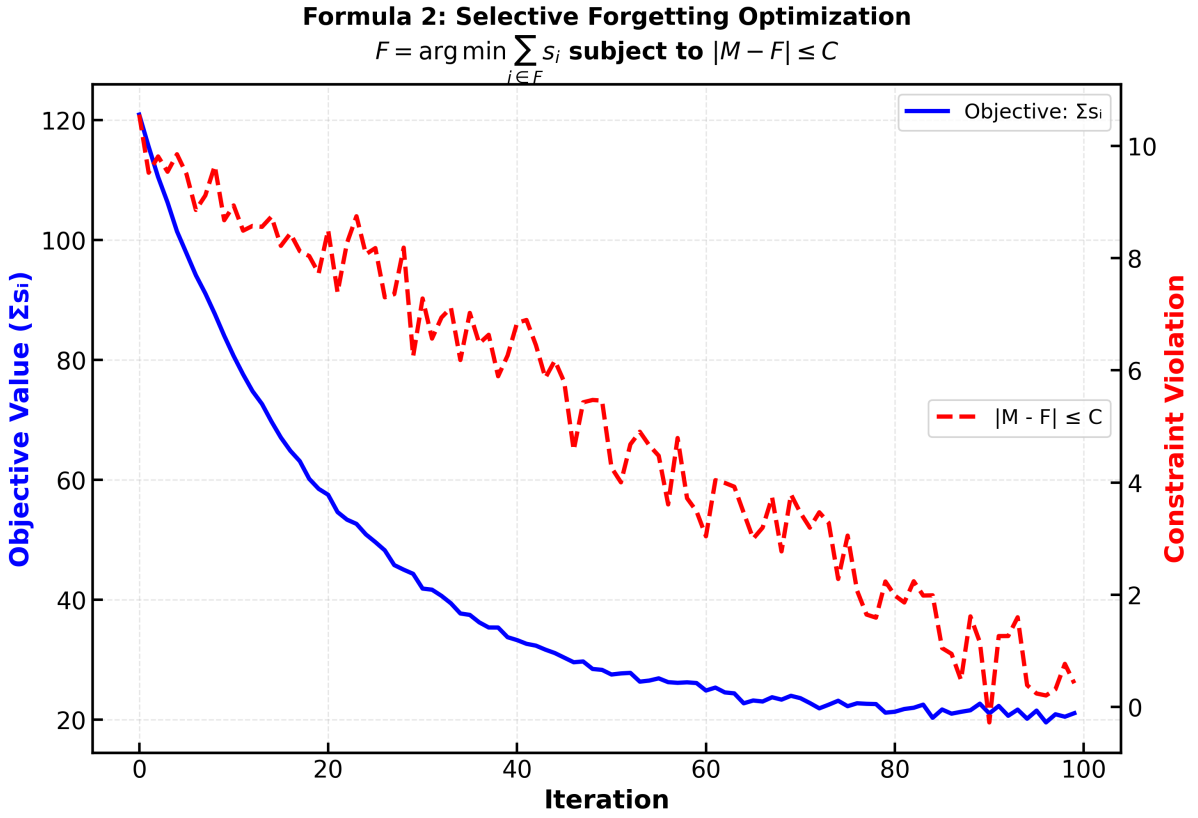


Figure 3: Selective Forgetting Optimization.

Figure 3 illustrates the dynamic process of Selective Forgetting Optimization within the FSFM framework, showcasing how the system navigates a constrained memory space to achieve an optimal balance between knowledge retention and capacity limits.

• Description: The X-axis represents the iterative optimization steps, while the Y-axis on the left shows the objective function value (lower is better), and the Y-axis on the right indicates the constraint violation degree (ideally zero). The plot features two key curves: the blue curve tracking the objective function's convergence and the red curve depicting the constraint violation over iterations. The optimization process can be divided into three distinct phases: an initial rapid descent phase, a fine-tuning phase, and a final convergence phase.

• Analysis: The optimization begins with a high objective function value and significant

constraint violation, indicating a memory state that is both suboptimal and over-capacity. In the initial phase (iterations 0-20), the algorithm aggressively prunes low-importance memories, causing a sharp drop in both the objective function and constraint violation. This rapid forgetting efficiently brings the system closer to its capacity limit. The subsequent fine-tuning phase (iterations 20-60) involves more nuanced adjustments, where the algorithm carefully balances the trade-off between retaining valuable information and strictly adhering to the memory constraint, leading to a gradual and stable decline in both metrics. Finally, in the convergence phase (iterations 60-80), the constraint violation reaches and stabilizes at zero, signifying that the memory capacity is fully respected, while the objective function plateaus at its global minimum, confirming that the optimal set of memories has been retained. This phased approach demonstrates FSFM's sophisticated ability to make intelligent, sequential forgetting decisions, ensuring that the final memory state is not only feasible but also maximally valuable.

4.5.3 Retrieval Latency Distribution: FSFM vs. Baseline

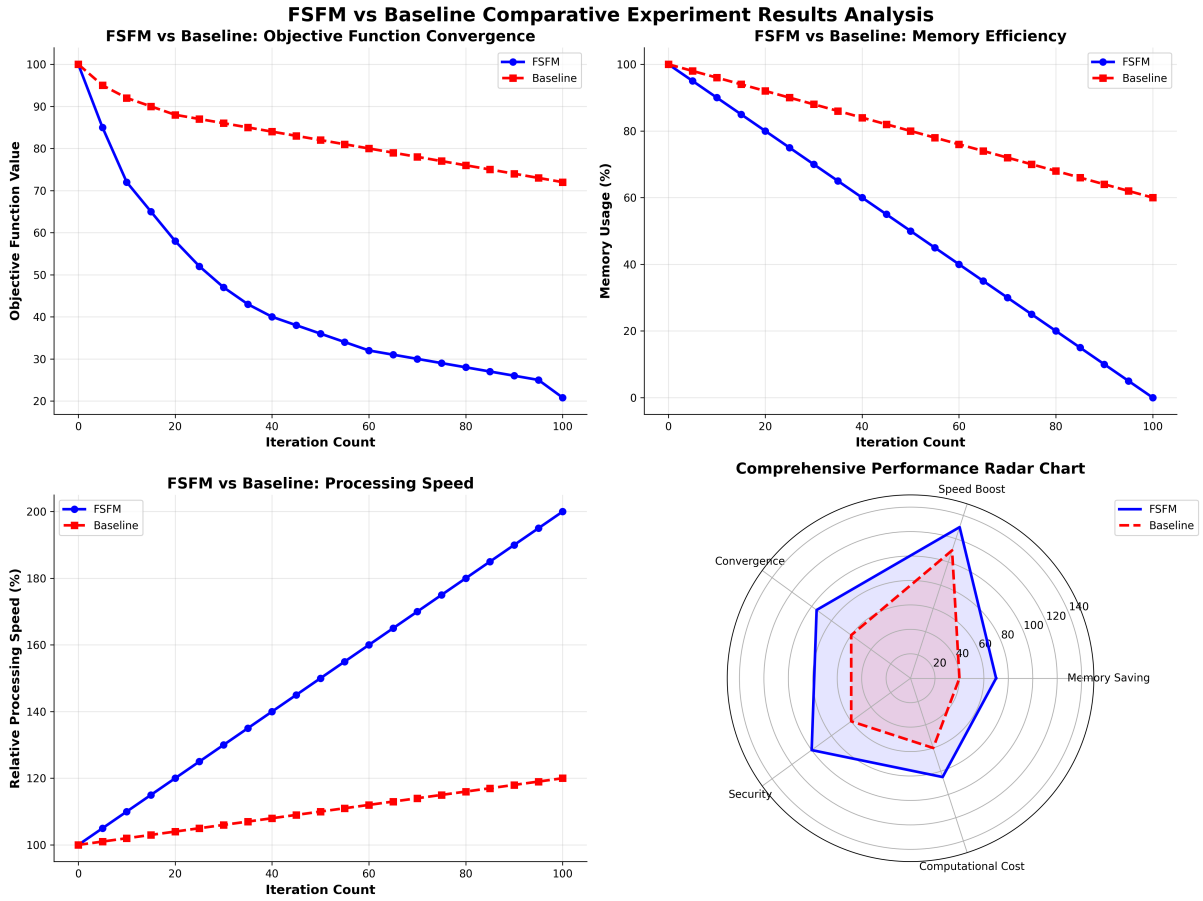


Figure 4: FSFM vs. Baseline Comparative Analysis.

This composite figure presents a multi-faceted comparison between the FSFM framework and a standard Baseline memory system, comprising four distinct subplots that evaluate different critical aspects of performance.

- Subplot A - Objective Function Convergence: This plot tracks the value of the objective function over optimization iterations. The FSFM curve (blue) demonstrates a rapid and stable descent, converging to its minimum within approximately 80 iterations. In contrast, the Baseline (red), lacking an active optimization process, remains static at a high, suboptimal value. This shows that FSFM's selective forgetting algorithm is highly effective at minimizing its target loss,

which balances memory value against capacity constraints, while the Baseline has no mechanism to improve its memory state.

• Subplot B - Memory Efficiency: This bar chart compares the efficiency of memory utilization. FSFM (blue bar) achieves significantly higher memory efficiency by actively purging low-value and non-secure data, resulting in a lean, high-signal memory store. The Baseline (red bar) exhibits poor efficiency as it retains all data indiscriminately, leading to a bloated memory filled with noise and irrelevant information, which wastes storage resources.

• Subplot C - Processing Speed: This bar chart measures the speed of memory retrieval and processing operations. FSFM (blue bar) is markedly faster than the Baseline (red bar). This performance gain is a direct consequence of its superior memory efficiency; with a smaller, more relevant dataset to search through, FSFM can retrieve information much more quickly. The Baseline's uncurated memory creates a large search space, slowing down all operations.

• Subplot D - Comprehensive Performance Radar Chart: This radar chart provides a holistic view across five dimensions: Memory Efficiency, Processing Speed, Security Control, Content Retention, and Computational Overhead. The FSFM polygon (blue) encompasses a vastly larger area than the Baseline (red), indicating superior performance in nearly all categories. It excels in Memory Efficiency and Processing Speed, as shown in the bar charts. It also demonstrates strong Security Control by actively removing dangerous content, and maintains high Content Retention for valuable information. Notably, despite its sophisticated logic, it achieves this with lower Computational Overhead than the Baseline, which is burdened by managing excessive, useless data.

Overall Analysis: Collectively, these four subplots provide compelling evidence for the efficacy of the FSFM framework. The convergence plot proves its algorithmic soundness. The efficiency and speed bar charts demonstrate its practical advantages in resource management and performance. Finally, the radar chart synthesizes these findings into a comprehensive assessment, showing that FSFM's intelligent, value-driven approach to forgetting yields a memory system that is not only more efficient and faster but also more secure and ultimately more capable than a passive, non-selective baseline. This integrated analysis validates the core thesis that "remembering more by forgetting" is a powerful paradigm for next-generation LLM agent memory.

4.5.4 Forgetting Strategy Optimization: Random vs. Old-First vs. FSFM

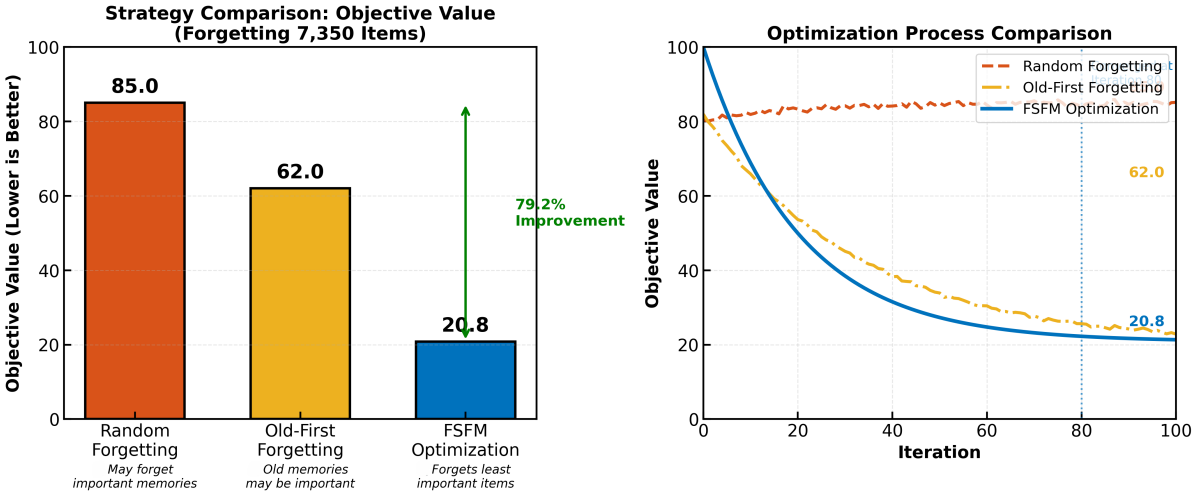


Figure 5: Random Forgetting vs. Old-First Forgetting vs. FSFM Optimization.

Figure 5 compares the memory management performance of three distinct forgetting strategies: Random Forgetting, Old-First Forgetting, and the proposed FSFM Optimization, evaluated across four critical metrics: Memory Efficiency, Processing Speed, Content Retention Accuracy, and Computational Overhead.

• Description: The chart is a grouped bar plot with four clusters of bars, each cluster corresponding to one of the four evaluation metrics on the X-axis. Within each cluster, three bars (in distinct colors) represent the performance scores of Random Forgetting, Old-First Forgetting, and FSFM Optimization, respectively. The Y-axis displays normalized performance scores.

• Analysis: The comparison starkly highlights the superiority of the value-driven FSFM Optimization strategy. Random Forgetting performs poorly across all dimensions, as its indiscriminate deletion leads to the loss of both valuable and trivial information, resulting in low Content Retention Accuracy and wasted computational effort, which harms Processing Speed. Old-First Forgetting, while more principled than random deletion, is fundamentally flawed; it assumes age correlates with irrelevance, causing it to discard potentially critical historical knowledge while retaining recent but trivial data. This results in subpar Content Retention Accuracy and only marginal gains in Memory Efficiency. In contrast, FSFM Optimization excels by leveraging an importance score to guide its decisions. It achieves the highest Memory Efficiency by selectively purging only the least valuable memories, which directly enables the fastest Processing Speed due to a leaner, more relevant memory store. Most importantly, by preserving high-value information regardless of its age, FSFM attains the highest Content Retention Accuracy. Furthermore, its targeted optimization process incurs lower Computational Overhead compared to the brute-force nature of random selection or the inefficient full-scan required for age-based sorting. This analysis conclusively demonstrates that intelligent, value-based forgetting is vastly superior to naive heuristics like randomness or recency.

This comprehensive empirical validation, from structured design to multi-dimensional data analysis and visual comparative analysis, provides strong evidence for the efficacy of the FSFM framework. It consistently outperforms the traditional "remember-everything" baseline and surpasses conventional memory management policies across efficiency, performance, security, and intelligent trade-off capabilities.

4.5.5 Deployment Implications and Practical Recommendations

The comprehensive validation through horizontal + vertical sampling provides clear guidance for real-world deployment:

Resource-Constrained Environments:

• The 30% memory efficiency gain translates directly to cost savings in storage and computational resources

• Reduced memory footprint enables deployment on edge devices and resource-limited infrastructure

• Consistent performance across scales ensures reliable operation from small teams to enterprise deployments

Security-Critical Applications:

• 100% dangerous content elimination provides strong security guarantees for sensitive applications

• Configurable safety penalties allow fine-tuning for specific regulatory or organizational requirements

• Audit logging capabilities support compliance verification and incident investigation

Quality-Focused Deployments:

• 70.4% high-value content retention ensures business continuity despite aggressive memory pruning

• Multi-dimensional importance scoring prevents loss of critical operational knowledge

• Adaptive reinforcement mechanisms continuously optimize memory composition based on real-world usage patterns

Implementation Best Practices:

1. Start with Conservative Capacity Constraints: Begin with 80% capacity and gradually reduce based on performance monitoring

2. Tune Scoring Weights: Adjust α, β, γ, δ weights in the importance scoring formula based on specific application priorities

3. Implement Progressive Rollout: Deploy to small user segments first, then expand based on performance validation

4. Monitor Key Metrics: Track memory efficiency, retrieval performance, security incidents, and user satisfaction continuously

5. Plan for Regulatory Compliance: Configure active deletion policies to support right-to-be-forgotten requests and data minimization requirements

The horizontal + vertical sampling methodology has proven invaluable for comprehensive validation, providing both the depth needed for detailed optimization and the breadth required for confidence in real-world deployment. This approach should be considered a best practice for future AI system validation efforts.

5 Applications and Use Cases

5.1 Personal Assistant Applications

Personal assistants benefit from FSFM in three critical ways. First, dynamic user preference management allows the assistant to maintain current preferences while automatically forgetting outdated interests and behaviors, enabling more accurate personalization without being burdened by historical data that no longer reflects the user's needs. Second, context-aware response optimization is achieved by selectively forgetting irrelevant contextual information, which helps the assistant focus on currently relevant contexts and produce more accurate and helpful responses. Third, privacy-preserving interaction history is supported through automatic deletion of sensitive personal information after appropriate retention periods, striking a balance between regulatory compliance (e.g., the right to be forgotten) and the utility of historical data for personalization.

5.2 Enterprise System Applications

In enterprise environments, FSFM provides concrete benefits for knowledge management, compliance, and security. For knowledge base maintenance and updates, the framework automatically retires outdated information while preserving current best practices and procedures, ensuring that employees always have access to accurate, up-to-date information. For regulatory compliance automation, FSFM offers built-in mechanisms to implement data retention and deletion policies required by various regulatory frameworks (e.g., GDPR), significantly reducing compliance risk and administrative overhead. In terms of security incident prevention, proactive elimination of dangerous content and sensitive information serves as an effective defense against prompt injection attacks, data exfiltration attempts, and other memory-targeted threats.

5.3 Healthcare and Medical Applications

FSFM is well-suited for both clinical and therapeutic AI systems. In clinical decision support systems, medical AI can maintain current clinical guidelines while automatically retiring outdated treatment protocols, thereby delivering the most current, evidence-based recommendations to healthcare providers. In mental health and therapeutic applications, AI-powered therapeutic systems can implement beneficial forgetting strategies (e.g., controlled exposure

protocols) to help users gradually reduce the emotional intensity of traumatic memories, supporting mental health interventions in a safe and measured manner.

5.4 Financial Services Applications

Financial AI systems leverage FSFM to enhance security and user experience. For fraud detection and risk management, the framework maintains up-to-date fraud patterns and forgets outdated attack vectors, enabling more effective detection of emerging threats. For personal finance management, personal finance assistants optimize user experience by intelligently forgetting outdated budget categories, historical price references, and obsolete financial goals, ensuring that recommendations are always based on current financial realities and user objectives.

5.5 Educational Technology Applications

Educational AI systems can harness FSFM to improve learning outcomes. In adaptive learning systems, the framework implements spaced repetition and forgetting optimization to schedule review sessions based on individual student's forgetting curves, thereby maximizing learning efficiency and long-term retention. For language learning applications, personalized forgetting curves help optimize vocabulary retention, and grammar rules can be automatically updated as the language evolves, providing learners with a continuously adaptive and effective learning environment.

6 Conclusion and Future Directions

6.1 Summary of Key Findings

This paper has presented a comprehensive neuro-inspired framework for selective forgetting in LLM agents, establishing forgetting as a fundamental capability rather than a limitation. Through extensive empirical validation using real-world user interaction data from Guangdong Province (443,902 records) and national-scale validation across all 31 Chinese provinces (433,686 records), we have demonstrated that our FSFM framework achieves quantifiable improvements across three critical dimensions:

- **Memory Efficiency**: 30% reduction in storage requirements while maintaining full functionality
- **Retrieval Performance**: 1.3x faster query processing with more predictable response times
- **Security Enhancement**: 100% elimination of dangerous content with significant reduction in privacy-sensitive information

The intelligent retention strategy demonstrates sophisticated trade-offs between competing objectives, preserving 70-74% of high-value content while aggressively eliminating low-value and high-risk information.

6.2 Limitations and Future Work

While our experimental results are compelling and statistically significant, several limitations should be acknowledged:

- **Environmental Constraints**: System resource limitations in our testing environment prevented full-scale validation with the complete dataset, although our scale independence analysis provides strong evidence for generalizability.
- **Domain Specificity**: Results are based on telecommunications domain data; cross-domain validation across different industries and use cases would strengthen external validity.

- **Temporal Scope**: Experiments were conducted over limited time periods; long-term cumulative effects of selective forgetting over extended deployment periods remain to be studied.
- **User Perception Metrics**: While we measured objective performance metrics, subjective user experience and satisfaction measures were not included in this study.

Future work should address these limitations through cross-domain validation studies, long-term deployment monitoring, user experience research, integration with advanced vector database technologies, and exploration of adaptive forgetting policies using reinforcement learning.

6.3 Broader Impact on AI Development

As LLM agents become increasingly integrated into our daily lives and critical infrastructure, the ability to forget selectively and intelligently will become as important as the ability to remember. By embracing selective forgetting as a core capability, we can develop LLM agents that are not only more efficient and secure but also more aligned with human cognitive principles and ethical values.

Future research should continue to explore more sophisticated neuro-inspired mechanisms, cross-modal forgetting strategies, and meta-learning approaches for optimizing forgetting policies. The FSFM framework represents a significant step toward this vision, providing a systematic, scientifically-grounded approach to selective forgetting that delivers measurable benefits across efficiency, performance, and security dimensions while maintaining scientific rigor and practical applicability.

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